Penokean remagnetization of the basal Huronian Dollyberry Lake basalts near Elliot Lake, Ontario

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The Dollyberry Lake volcanics occur at the base of the Huronian Supergroup and outcrop ~15 km north of Elliot Lake, Ontario. Ten specimens were collected from each of 36 sites in the volcanics and from a further five sites in stratigraphically equivalent volcanics ~9 km east of Elliot Lake. Alternating field (AF) and thermal step demagnetizations were done on two specimens from each site up to 100 mT and 650°C, respectively. AF demagnetization led to the isolation of an A component in 33 sites, a B component in three sites, and both components in two sites. Fold tests demonstrate that both the A and B components are postfolding in origin at the >97 and >80% confidence level, respectively. It is argued that the A component (pole 154°W, 56°N, $\delta p = 8°$, $\delta m = 10°$) was acquired during the Penokean Orogeny at ~2215 ± 40 Ma, and that the B component (pole 60°W, 20°S, $\delta p = 8°$, $\delta m = 13°$) was acquired probably at ~1900 ± 100 Ma in response to some geographically restricted thermal or chemical event. The results indicate that the coeval Thessalon volcanics studied by Symons and O'Leary retain the Penokean A component. This leads to the retraction of their conclusion that the superadjacent uranium ores were formed at an ~35° paleolatitude and to the conclusion by analogy that the Penokean Orogeny was the only significant orogenic event in the evolution of the ore deposits.

Les roches volcaniques du lac Dollyberry se situent à la base du supergroupe de l'Hudsonien et affleurent à ~15 km au nord du lac Elliot, Ontario. Dix échantillons furent prélevés à chacun des 36 sites dans des roches volcaniques et à cinq sites supplémentaires dans des roches volcaniques stratigraphiquement équivalentes à ~9 km à l'est du lac Elliot. La désaimantation par champs alternatifs (CA) et par traitements thermiques a été effectuée sur deux échantillons pour chacun des sites jusqu'à 100 mT et 650°C respectivement. La désaimantation CA permet de résoudre une composante A pour 33 sites, une composante B pour trois sites et les deux composantes pour deux sites. Des essais tenant compte de plissement démontrent que les deux composantes A et B ont une origine postérieure au plissement avec un niveau de confiance >97 et >80% respectivement. On croit que la composante A (pôle 154°O, 56°N, $\delta p = 8°$, $\delta m = 10°$) fut acquise durant l'orogénèse du Pénokéen il y a ~2215 ± 40 Ma, et que la composante B (pôle 60°O, 20°S, $\delta p = 8°$, $\delta m = 13$) fut acquise probablement il y a ~1900 ± 100 Ma par reaction à un évènement thermique ou chimique géographiquement restreint. Les résultats indiquent que les roches volcaniques contemporaines du Thessalon étudiées par Symons et O'Leary retiennent la composante A pénokéenne. Ces observations nécessitent une révision de leur conclusion stipulant que les minerais d'uranium supra-adjacents furent formés à une paléolatitude de ~35° et de la conclusion déduite par analogie que l'orogénèse du Pénokéen fut le seul évènement orogénique important affectant l'énvolution des dépôts de minerals.

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Introduction

Volcanic rocks are found at or near the base of the Huronian Supergroup in several places throughout the Penokean fold belt of the Precambrian Southern Tectonic Province (Fig. 1). One occurrence, named the Dollyberry Lake volcanics by Robertson (1969), outcrops along the north limb of the Quirke syncline to the northwest of Elliot Lake, Ontario. Paleomagnetic study of the volcanics is of interest for two reasons. First, Robertson (1973) has pointed out the close spatial relationship between the volcanics and the important uranium ore deposits of the area. Thus paleomagnetic information on the age and evolution of the volcanics also yields evidence on the ore genesis. Second, Symons and O'Leary (1978) found that the contemporaneous flat-lying Thessalon volcanics have an ~2375 ± 75 Ma pole position. This fits the known geologic relationships but conflicts with their ~2018 ± 49 Ma Rb–Sr isochron age. The steeply dipping Dollyberry Lake volcanics provide the opportunity for a statistically significant fold test to determine if these volcanics retain a prefolding remanence rather than a Penokean or later orogenic remagnetization.

Geology

The Huronian Supergroup nonconformably overlies Archean granitic rocks and metastrata that yield isotopic ages ranging from 2.40 to 2.75 Ga (Roscoe 1973). Except for the basal volcanics, the Huronian comprises ~10 000 m of clastic sediments with minor chemical sedimentary units. The volcanics occur in the lowermost Elliot Lake Group where they conformably overlie either a paleosol resting on the Archean basement (Robertson 1963, 1968) or the Livingstone Creek subarkose formation (Frarey 1977).

The Dollyberry Lake volcanics are basaltic in compo-

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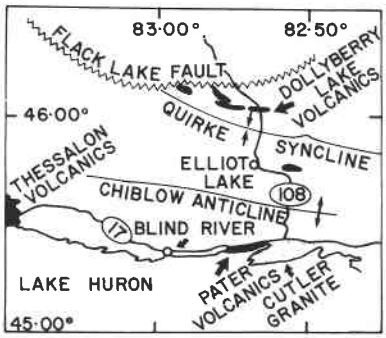


FIG. 1. Location map of the northshore of Lake Huron.

sition with amygdaloidal and (or) porphyritic zones (Robertson 1973). They are unmetamorphosed or weakly metamorphosed—low greenschist facies at most—and characteristically contain either disseminated or amygdule-infilling chalcopyrite mineralization. The fine-grained flow tops and interflow tuffs are often sheared. It is probable that the 2215 ± 185 Ma Rb–Sr isochron reported by Knight and Fairbairn (1966) represents the Dollyberry Lake volcanics.

Several basic volcanic units are correlated with the Dollyberry Lake basalts. An unnamed occurrence to the east of Elliot Lake on the south limb of the Quirke syncline (sites 1–5, Fig. 2) was originally thought to be Archean (Robertson 1961, 1973). The Thessalon volcanics outcrop in several places to the southwest and west (Frarey 1977) (Fig. 1). Five Rb–Sr whole-rock isochron ages from unmetamorphosed basic to acidic flow units give a combined isochron of 2018 ± 49 Ma for these volcanics (Van Schmus 1965; Knight and Fairbairn 1966; Fairbairn *et al.* 1969). One from metamorphosed basalt in Aberdeen Township gives an age of 1797 ± 24 Ma. The Pater mafic volcanics outcrop to the south on the southern limb of the Chiblow anticline and have been metamorphosed to amphibolite grade. They were originally mapped as Archean or Archean (?) (Robertson 1970), but are now considered to be basal Huronian (Robertson 1973). Knight and Fairbairn (1966) obtained a Rb–Sr whole-rock isochron of 2496 ± 145 Ma for these flows. In the Espanola–Sudbury area to the east, the Stobie basic volcanics and overlying Copper Cliff acidic volcanics are also considered coeval (Roscoe 1973). The Stobie volcanics give a Rb–Sr whole-rock isochron of 2005 ± 125 Ma (Fairbairn *et al.* 1969); however, the overlying Copper Cliff volcanics give two Rb–Sr whole-rock isochrons of 2350 Ma (Fairbairn 1965).

The Dollyberry Lake basalts are conformably overlain by uraniferous oligomictic conglomerates and gritty subarkose of the Matinenda Formation of the Elliot Lake Group, and above these are the rest of the Huronian Supergroup. Unmetamorphosed sediments from the

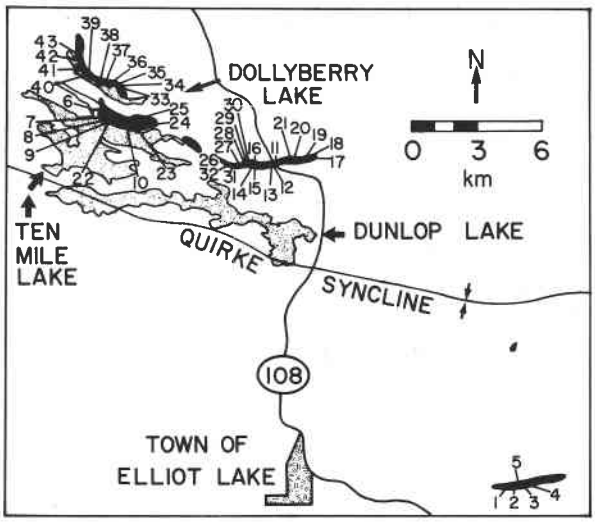


FIG. 2. Site location map for the Dollyberry Lake volcanics.

upper Huronian Gowganda Formation near Gowganda, Ontario give a well defined Rb–Sr whole-rock isochron of 2288 ± 87 Ma (Fairbairn *et al.* 1969). Rb–Sr whole-rock isochrons from metamorphosed mid-Huronian formations of the greenschist and amphibolite facies from the Espanola–Sudbury area give values of 1802 ± 105, 1810 ± 90, 2042 ± 48, and 2174 ± 160 Ma with a combined average of 1950 ± 100 Ma (Fairbairn *et al.* 1969).

Negative paleomagnetic fold tests on the Nipissing diabase in several areas of the Southern Province (Symons 1970, 1971; Symons and Londry 1975; Stupavsky and Symons 1982*a*) indicate that the Huronian strata were essentially deformed into their present "basin and arch" or "syncline–anticline" structural pattern during the Penokean Orogeny (Church 1968) or prior to Nipissing intrusion. Nipissing diabase dikes, sheets, and sills were emplaced extensively into the Archean and Huronian units, and are known to cut the Dollyberry Lake basalts in several places. The nearby Mount Lake "diabase" differentiated from diorite to granophyre, gives a Rb–Sr whole-rock age of 2130 ± 215 Ma (Van Schmus 1965; Hofmann 1967). In addition, the Nipissing diabase has yielded two well defined Rb–Sr whole-rock isochrons of 2155 ± 80 and 2162 ± 27 Ma (Van Schmus 1965; Fairbairn *et al.* 1969), as well as Rb–Sr and K–Ar biotite ages of 2180 ± 40 and 2095 Ma (Van Schmus 1965; Lowdon *et al.* 1963) Paleomagnetic work suggests the presence of two phases (N1 and N2) of Nipissing intrusion in the Penokean fold belt (Symons 1971; Morris 1979, 1981). These appear to be coincident with two phases (Ab1 and Ab2) of Abitibi dike intrusion that give Rb–Sr whole-rock isochrons of 2206 ± 132 and 2102 ± 67 Ma (Symons *et al.* 1981) when the original data of Gates and

TABLE 1. Mean AF and thermal step demagnetization data for the pilot specimens

Demag. field	A component (32 specimens)					B component (5 specimens)				
	D(°)	I(°)	J_n/J_0 Mean	SD	α_{95} (°)	D (°)	I (°)	J_n/J_0 Mean	SD	α_{95} (°)
AF (mT)										
0	308	70	1.00	0.00	13	324	−29	1.00	0.00	56
5	313	69	0.59	0.21	11	328	−29	0.74	0.35	43
10	312	68	0.40	0.25	10	333	−33	0.69	0.41	20
15	309	68	0.29	0.23	10	333	−42	0.68	0.41	10
20	311	65	0.22	0.19	10	333	−45	0.62	0.39	11
30	310	64	0.15	0.15	10	334	−45	0.52	0.33	16
40	305	69	0.12	0.12	12	331	−42	0.37	0.25	14
50	298	66	0.095	0.10	15	334	−39	0.26	0.18	19
60	310	63	0.083	0.089	18	334	−38	0.18	0.12	28
80	319	58	0.066	0.087	22	313	−44	0.091	0.072	60
100	291	68	0.054	0.073	21	356	22	0.042	0.023	85
Thermal (°C)										
0	321	68	1.00	0.00	9	312	−34	1.00	0.00	42
200	314	64	0.70	0.14	11	313	−46	1.02	0.16	36
400	311	55	0.31	0.15	14	313	−51	0.86	0.20	28
500	317	54	0.20	0.11	14	316	−53	0.82	0.25	27
550	315	56	0.14	0.10	17	257	−18	0.35	0.42	47
570	315	63	0.094	0.068	22	299	33	0.29	0.41	90
620	316	64	0.069	0.061	31	53	−44	0.035	0.033	90
650	295	63	0.059	0.060	33	105	17	0.015	0.010	67

NOTES: Six pilot specimens having outlier directions have been excluded from the means. J_n/J_0 = normalized remanence intensity.

Hurley (1973) are recalculated. Robertson (1973) and Frarey (1977) cite field evidence for two phases of diabase intrusion. The Nipissing diabase in the Elliot Lake area is unaltered or, at most, marginally altered to lowest greenschist metamorphic rank (Robertson 1968; Card and Pattison 1973) indicating minimal subsequent metamorphic activity.

Evidence for post-Nipissing metamorphism around Dollyberry Lake is minimal also. The 1750 Ma Hudsonian Orogeny may be represented by a faulting event; however, the closest intrusive unit is the Cutler granite, which is some 50 km away (Robertson 1973). In the immediate area are occasional olivine diabase dikes of the ~1225 Ma Sudbury swarm. There is no evidence for Grenvillian or subsequent orogenic events.

Sampling and measurement

Five cores per site were drilled at 38 sites in the Dollyberry Lake basalts and at five sites in the equivalent volcanics to the east of Elliot Lake (Fig. 2). The cores were oriented *in situ* using a sun compass or a magnetic compass with topographic check sitings. The sites were distributed both horizontally and vertically to sample the exposed stratigraphic section fairly uniformly. Two or more 2.4 cm right cyclindrical specimens were sliced from each core. The natural remanent magnetization (NRM) was measured using a Schonstedt SSM-1A spinner magnetometer. Two pilot specimens with typical NRM characteristics were selected from one core for each site. One pilot specimen was alternating field (AF) demagnetized in 11 steps up to 100 mT and the second pilot specimen was thermally demagnetized in eight steps up to 650°C using Schonstedt GSD-1 and TSD-1 demagnetizers, respectively. A further eight specimens, two per core from each site, were AF demagnetized at 20 mT.

Step demagnetization

AF and thermal step demagnetization of the 86 pilot specimens isolates A and B magnetization components.

The dominant A component is centred at ~320°, 65° (declination, inclination) and is found in 32 specimens (Table 1; Fig. 3). The A remanence carriers have a broad coercivity spectrum with most release forces being <20 mT and unblocking temperatures being <500°C. However, the A component survives in most specimens up to 50 mT and 570°C and in a few specimens up to 100 mT and 650°C. The A component resides, therefore, in both coarse- and fine-grained magnetite and in hematite.

The minor B component is centred at ~330°, −40° and is found in only five AF and five thermal pilot

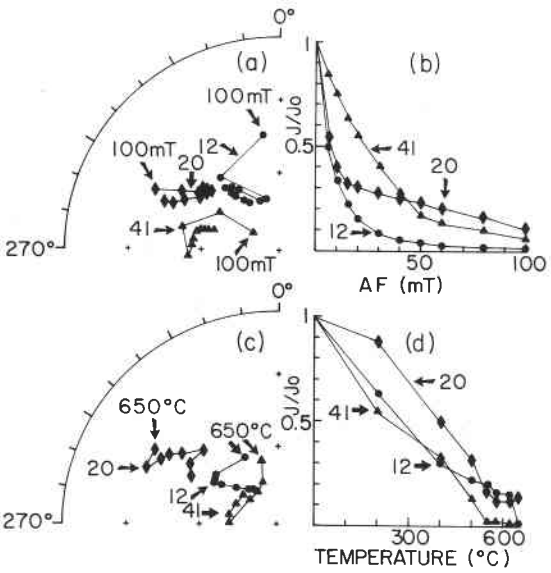


FIG. 3. Example step demagnetization data for the A component. The pilot specimens represent sites 12, 20, and 41. The directions are plotted in the lower hemisphere of a Schmidt equal-area stereonet segment. J/J_0 is the ratio of the measured remanence intensity to the NRM intensity. The AF cleaning field is in milliteslas (mT) and the thermal cleaning temperature in degrees centigrade.

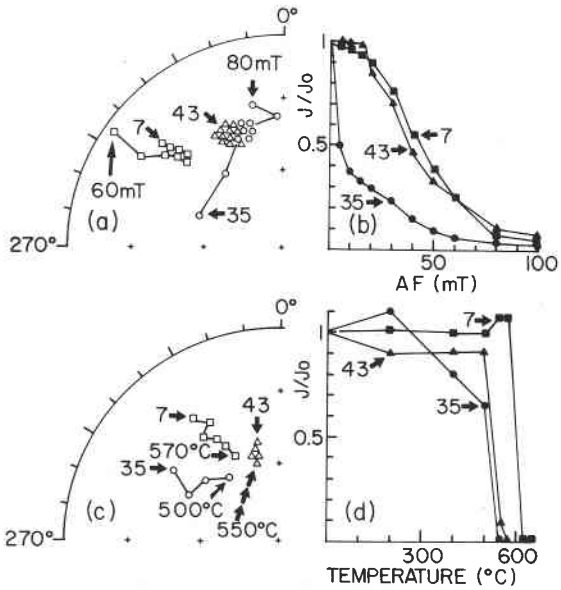


FIG. 4. Example step demagnetization data for the B component. Conventions are as given in Fig. 3 except that the directions are plotted in the upper hemisphere of the Schmidt stereonet.

specimens (Table 1; Fig. 4). The B remanence carriers have a narrow coercivity spectrum with most release forces being >20 mT and unblocking temperatures being >500°C. The B component survives only to 60 mT and 570°C, and hence appears to reside in fine-grained magnetite only.

Site mean remanence

After 20 mT AF demagnetization, the remanence data exhibit relatively good within-specimen, within-site, and within-rock unit remanence homogeneity (Tables 2, 3). Specimen level screening (Harrison 1980; Briden and Arthur 1981; Stupavsky and Symons 1982*b*) led to the rejection of 14.1% of the remanence directions that were derived from inhomogeneously magnetized specimens. After rejecting a further 7.7% of the specimen directions as outliers, site mean directions with cones of 95% confidence (α_{95}) of <25° were found for all but four sites (Table 1). The A component only was isolated at 33 sites, the B component only at three sites, the A component was reversed at one site (39), and the A and B components together were found at two sites (42, 43).

Fold tests

The Dollyberry Lake basalts (sites 6–43) were examined initially using the paleomagnetic fold test of Graham (1949). Using the 11 (31) site means with an $\alpha_{95} \leq 10°$ (25°), the test shows that the basalts acquired the A component after folding at the >99% (>97%) confidence level. Using all five site means, the test indicates that the B component was also acquired after folding with >80% confidence.

The "south limb" volcanics (sites 1–5) are flat lying. They have a mean remanence direction that is just barely statistically different at the 95% confidence level from the uncorrected Dollyberry Lake basalts A direction, and markedly different from the corrected A direction. Thus it appears that, despite being flat lying, these volcanics also retain a postfolding metamorphic remanence.

Similarly, the gently dipping Thessalon volcanics also yield an uncorrected mean direction that is relatively close to although statistically different from the uncorrected Dollyberry Lake A component direction and markedly different from the corrected A direction. Recognizing that the fold test on the Thessalon volcanics (Symons and O'Leary 1978) was inconclusive, this implies that their remanence is also postfolding in origin.

The fact that the uncorrected directions from the Dollyberry Lake A, south limb A, and Thessalon volcanics are just statistically significantly different at the 95% confidence level may be explained by a variety of mechanisms including the following: (a) secular variation is not completely averaged out especially for the south limb volcanics; (b) minor postmetamorphic rotations have occurred between the outcrop areas; (c) a minor residual primary remanence is included with the

TABLE 2. Site mean remanence data after 20 mT AF demagnetization and before tilt correction

Site	Remanence component	N	N^+	N^*	D (°)	I (°)	R	α_{95} (°)	Strike (°)	Dip (°)
1	A	7	0	2	297	65	6.357	21	0	0
2	A	6	1	2	271	56	5.459	23	0	0
3	A	6	2	1	280	76	5.605	20	0	0
4	A	11	0	0	258	35	10.384	12	0	0
5	A	9	0	0	276	74	8.125	18	0	0
6	A	8	0	1	271	58	7.321	18	79	25
7	B	8	1	0	329	−32	7.360	13	120	35
8	A	7	2	0	294	26	6.656	15	120	35
9	A	5	0	4	334	48	4.781	18	120	35
10	A	8	1	0	321	52	7.598	14	120	35
11	A	7	1	1	354	66	6.978	4	70	35
12	A	9	0	0	331	63	8.829	8	70	35
13	A	7	0	2	282	69	6.599	16	70	35
14	A	8	1	0	309	68	7.900	7	110	50
15	A	7	1	1	290	76	6.911	7	110	50
16	A	7	1	1	326	74	6.754	12	110	50
17	A	8	1	0	280	64	7.694	12	70	35
18	A	4	2	3	303	15	3.687	31	70	35
19	A	7	0	2	300	60	6.694	14	70	35
20	A	7	0	2	325	81	4.810	45	70	35
21	A	8	1	0	307	64	7.890	7	70	35
22	A	7	0	2	20	41	4.643	48	120	35
23	A	6	0	3	354	26	5.595	20	120	35
24	A	6	0	3	2	55	5.337	26	120	35
25	A	6	0	3	283	52	5.618	19	120	35
26	A	6	1	2	43	85	5.676	18	110	50
27	A	9	0	0	338	70	8.846	7	110	50
28	A	7	1	1	326	70	6.812	11	110	50
29	A	7	1	1	18	72	6.911	7	110	50
30	A	9	0	0	335	71	8.829	8	110	50
31	A	8	0	1	324	78	7.798	10	110	50
32	A	7	2	0	317	72	6.895	8	110	50
33	A	5	2	2	325	57	4.835	16	120	75
34	A	6	1	2	312	41	5.540	21	120	75
35	B	8	1	0	343	−37	7.890	7	120	75
36	A	4	0	5	329	77	3.984	7	120	75
37	A	8	0	1	324	40	7.738	11	120	70
38	A	5	1	3	62	74	4.765	19	120	70
39	A, reversed	4	1	4	148	−40	3.956	11	120	70
40	B	6	3	0	332	−24	5.748	15	120	70
41	A	9	0	0	307	58	8.639	11	120	70
42A	A	5	0	0	307	35	4.771	19	120	70
42B	B	4	0	0	342	−47	3.994	4	120	70
43A	A	5	1	0	273	55	4.870	14	120	70
43B	B	5	0	0	347	−44	4.936	10	120	70

NOTES: N is the number of accepted specimens; N^+ is the number of specimens rejected as having outlier directions; N^* is the number of specimens rejected as being inhomogeneously magnetized, i.e., having within-specimen remanence angular standard deviation >10°; R is the length of vector resultant; α_{95} is the radius of circle of 95% confidence about the mean (Fisher 1953).

A component; and (d) the metamorphic remagnetization was not strictly contemporaneous for each outcrop area.

As a final check, when all the mean directions from the 55 sites with $\alpha_{95} \leq 25°$ from the three volcanic outcrop areas are subjected to the fold test, their A component remanence is postfolding in origin at the >99% confidence level.

Age of the A and B components

The postfolding A component of the Dollyberry Lake

TABLE 3. Unit mean remanence data and fold tests

Unit	N	SC α_{95} (°)	C	Uncorrected D (°)	Uncorrected I (°)	Uncorrected k	Uncorrected α_{95} (°)	Tilt-corrected D (°)	Tilt-corrected I (°)	Tilt-corrected k	Tilt-corrected α_{95} (°)	Fold test: Remanence acquired	Fold test: CL
Dollyberry Lake basalts	11	⩽10	A	329	72	81	5	216	55	17	12	postfolding	>99%
Dollyberry Lake basalts	31	⩽25	A	316	63	16	7	237	48	8	10	postfolding	>97%
Dollyberry Lake basalts	5	⩽15	B	338	−37	53	11	349	12	25	16	postfolding	>83%
"South limb" volcanics	5	⩽25	A	273	62	20	18	273	62	20	18	—	—
Thessalon volcanics*	19	⩽25	A	303	53	28	6	303	51	26	7	inconclusive	—
All volcanics	55	⩽25	A	307	60	17	5	262	55	7	8	postfolding	>99%

NOTES: N is the number of site means; SC is the selection criterion using α_{95} of Fisher (1953); C is the component averaged; k is the precision parameter of Fisher (1953); CL is the confidence level expressed as a percentage.
*Taken from Symons and O'Leary (1978).

basalts gives a pole position of 154°W, 56°N ($\delta p = 8°$, $\delta m = 10°$) (Fig. 5; Table 4). Following the conventional interpretation that the 2288 ± 87 Ma Rb–Sr isochron for the Gowganda sediments dates primary deposition, and allowing for both the deposition of the balance of the upper Huronian Supergroup and for postdepositional folding, the maximum age for the A remagnetization is ~2250 Ma. It is most likely that the A component predates intrusion of the Nipissing N1 intrusives because they were emplaced into folded Huronian strata and they retain an unreset primary remanence with radically different pole positions. The three relevant Rb–Sr isochrons of 2206 ± 132 Ma (Gates and Hurley 1973; Symons *et al.* 1981), 2162 ± 27 Ma (Fairbairn *et al.* 1969), and 2155 ± 80 Ma (Van Schmus 1965) set emplacement at ~2175 Ma.

Three features are attractive about the interpretation that the A component is a Penokean remagnetization set between ~2250 and ~2175 Ma. First, it fits neatly with the 2215 ± 185 Ma Rb–Sr isochron apparently given by the basalts (Knight and Fairbairn 1966), provided that the Rb–Sr decay system was reset also during the orogeny. Second, the A pole of the basalts is similar to poles from the Gowganda Formation, which have been shown to be postdeformational by fold tests and pre-Nipissing by contact tests (Symons 1967, 1975; Roy and Lapointe 1976; Morris 1977, 1979; Stupavsky and Symons 1982*a*). Third, the pole falls reasonably close to the ~2200 Ma position on recent apparent polar wander (APW) paths for North America (e.g., Irving 1979; Stupavsky and Symons 1982*a*).

The possibility that the A component of the Dollyberry Lake volcanics is related to some younger metamorphic event appears unlikely. The poles of the Nipissing N2 and Abitibi Ab2 dikes form a cluster that is remote from the A pole. Also few, if any, of the Dollyberry Lake sites are within about two dike widths of known Nipissing dikes. In the same vein, the A pole is remote from known positions of Hudsonian Orogeny, Sudbury swarm, and Grenville Orogeny poles so that correlation to these events appears untenable.

Given their similarity to the Dollyberry Lake A pole, the poles of the south limb A and Thessalon volcanics are also interpreted to be Penokean orogenic remagnetizations acquired ~2215 ± 40 Ma ago. This negates the interpretation of Symons and O'Leary (1978) that the Thessalon volcanics and overlying unraniferous quartz-pebble conglomerates were deposited at an ~35° paleolatitude. It does, however, indicate that post-Penokean tectonic and thermal effects on the ores were minimal.

Explanation of the Dollyberry Lake volcanics B component is more tenuous. It is probably postfolding in origin based on the fold test and on its limited coercivity

TABLE 4. Pole positions for Huronian volcanics

Unit	N	C	Pole position: Long. (°W)	Pole position: Lat. (°N)	Radii of 95% oval: δp (°)	Radii of 95% oval: δm (°)
Dollyberry Lake basalts	31	A	160	59	9	11
Dollyberry Lake basalts	5	B	61	−20	7	13
"South limb" volcanics	5	A	141	32	22	28
Thessalon volcanics	19	A	168	47	6	9
All volcanics	55	A	162	52	6	8

NOTES: *N* is the number of sites; C is the component; the pole position is given by its longitude west, latitude north (+ve), and the length of the semi-axes of the oval of 95% confidence perpendicular to (δp) and along (δm) the site pole great circle.

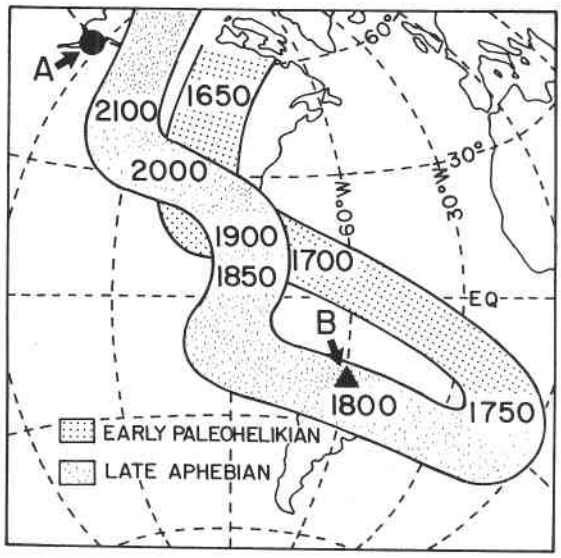


FIG. 5. Location of the poles for the A and B components plotted on the APW path of Irving (1979).

spectrum, which falls within the metamorphic A component spectrum. Its occurrence is restricted to the western end of the outcrop belt where it is found as the sole component at three sites and as a joint component with A at two sites. These sites range stratigraphically from top to bottom of the basalt unit, and they are remote from known later intrusives or major faults that cut the unit. The pole position of the B component pole, 60°W, 20°S (δp = 8°, δm = 13°), is located squarely on the 1800 and 2000 Ma positions of Irving's (1979) and Stupavsky and Symons' (1982*a*) versions of the APW path, respectively. It may therefore reflect a variety of minor Hudsonian orogenic events such as the intrusion of a minor set of northeast- or northwest-trending quartz diabase dikes or alteration by thermal fluids along one of the several sets of faults.

Conclusions

The Dollyberry Lake basalts retain two postfolding remanence components. The common A component was acquired during the Penokean Orogeny at ~2215 ± 40 Ma. At the same time, the "south limb" volcanics and the Thessalon volcanics acquired their postfolding metamorphic remanence. The uncommon B component is geographically very restricted and was probably acquired after folding at ~1900 ± 100 Ma. The Symons and O'Leary (1978) interpretation of an ~35° paleolatitude for deposition of the superjacent uraniferous ores is retracted. The Penokean Orogeny produced the bulk of the deformation and regional metamorphism affecting the basalts and, by analogy, the ore.

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